EXP.2. Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

Aim:
To identify the optimum operating frequency within the 8–12 GHz range by determining the frequency at which \(|S_{11}|\) is minimum and \(|S_{21}|\) is maximum, and compare its matching and transmission performance with that at 10 GHz.

Software required:

Matlab

Program:

```
clc;
clear;
close all;

%% PART B: OPTIMUM FREQUENCY ANALYSIS

%% Input Parameters
Z0 = 50;                    % Reference impedance (ohms)
Zc = 52;                    % Characteristic impedance (ohms)
L = 50e-3;                  % Transmission line length (50 mm)
c = 3e8;                    % Speed of light (m/s)
er = 2.2;                   % Relative permittivity
loss_dB_per_m = 10;          % Assumed attenuation (dB/m)

%% Frequency sweep from 8 GHz to 12 GHz
f = linspace(8e9, 12e9, 1001);

%% Calculate Transmission Line Parameters
beta = 2*pi*f*sqrt(er)/c;
alpha = loss_dB_per_m/8.686;
gamma = alpha + 1j*beta;

%% ABCD Parameters
A = cosh(gamma*L);
B = Zc*sinh(gamma*L);
C = sinh(gamma*L)/Zc;
D = cosh(gamma*L);

%% Calculate S-Parameters
den = A + B/Z0 + C*Z0 + D;

S11 = (A + B/Z0 - C*Z0 - D)./den;
S21 = 2./den;

%% Find Optimum Frequencies
[~, idx_min] = min(abs(S11));
[~, idx_max] = max(abs(S21));

f_best_S11 = f(idx_min);
f_best_S21 = f(idx_max);

fprintf('\nAt optimum S11 frequency:\n');
fprintf('|S11| = %.6f\n', abs(S11(idx_min)));
fprintf('|S21| = %.6f\n', abs(S21(idx_min)));
fprintf('Return Loss = %.4f dB\n', RL_best);
fprintf('Insertion Loss = %.4f dB\n', IL_best);

fprintf('\nAt 10 GHz:\n');
fprintf('Return Loss = %.4f dB\n', RL_10);
fprintf('Insertion Loss = %.4f dB\n', IL_10);

%% Plot 1: S11 versus Frequency
figure;
plot(f/1e9, 20*log10(abs(S11)), 'b', 'LineWidth', 1.5);
hold on;
plot(f_best_S11/1e9, ...
    20*log10(abs(S11(idx_min))), ...
    'ro', 'MarkerFaceColor', 'r');

xlabel('Frequency (GHz)');
ylabel('|S11| (dB)');
title('S11 versus Frequency');
legend('S11', 'Minimum |S11|');
grid on;

%% Plot 2: S21 versus Frequency
figure;
plot(f/1e9, 20*log10(abs(S21)), 'r', 'LineWidth', 1.5);
hold on;
plot(f_best_S21/1e9, ...
    20*log10(abs(S21(idx_max))), ...
    'bo', 'MarkerFaceColor', 'b');

xlabel('Frequency (GHz)');
ylabel('|S21| (dB)');
title('S21 versus Frequency');
legend('S21', 'Maximum |S21|');
grid on;

%% Plot 3: Return Loss versus Frequency
figure;
```

```
%% Return Loss and Insertion Loss at Optimum S11 Frequency
RL_best = -20*log10(abs(S11(idx_min)));
IL_best = -20*log10(abs(S21(idx_min)));

%% Results at 10 GHz
[~, idx10] = min(abs(f - 10e9));

RL_10 = -20*log10(abs(S11(idx10)));
IL_10 = -20*log10(abs(S21(idx10)));

%% Display Results
fprintf('\nPART B: OPTIMUM FREQUENCY ANALYSIS\n');
fprintf('================================\n');

fprintf('Minimum |S11| frequency = %.4f GHz\n', ...
    f_best_S11/1e9);

fprintf('Maximum |S21| frequency = %.4f GHz\n', ...
    f_best_S21/1e9);
```

```
plot(f/1e9, -20*log10(abs(S11)), 'LineWidth', 1.5);
xlabel('Frequency (GHz)');
ylabel('Return Loss (dB)');
title('Return Loss versus Frequency');
grid on;

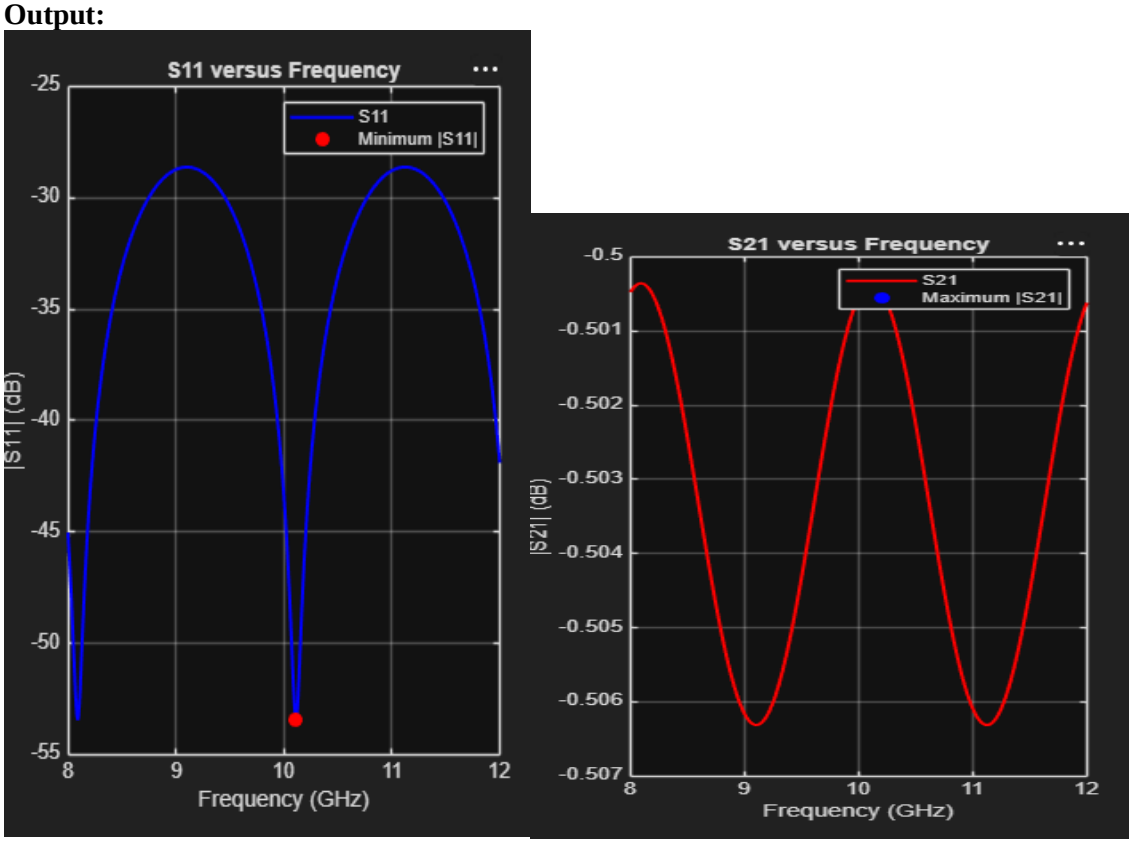
%% Plot 4: Insertion Loss versus Frequency
figure;
plot(f/1e9, -20*log10(abs(S21)), 'LineWidth', 1.5);
xlabel('Frequency (GHz)');
ylabel('Insertion Loss (dB)');
title('Insertion Loss versus Frequency');
grid on;

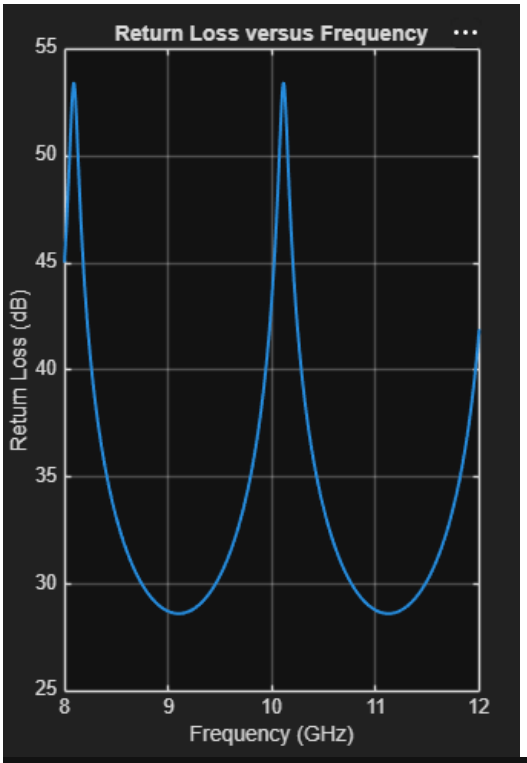
%% Compare Optimum Frequency with 10 GHz
fprintf('\nFREQUENCY COMPARISON\n');
fprintf('====================\n');
fprintf('Reference frequency = 10 GHz\n');

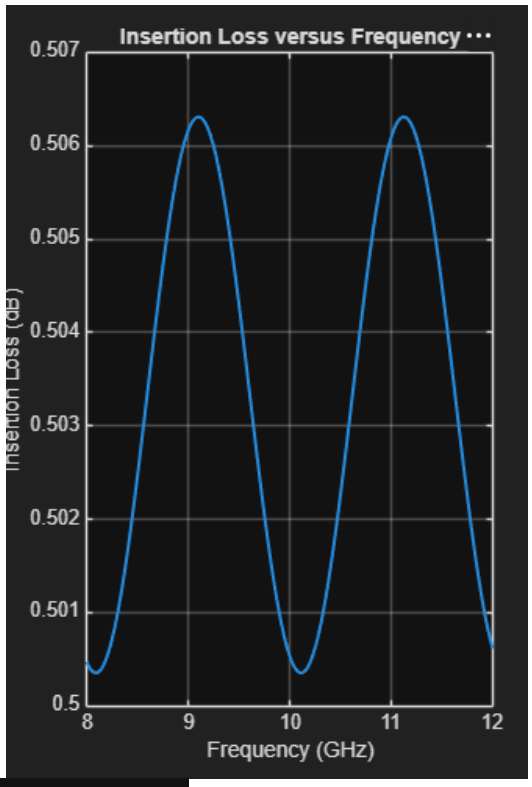
fprintf('Optimum S11 frequency = %.4f GHz\n', ...
    f_best_S11/1e9);

fprintf('Optimum S21 frequency = %.4f GHz\n', ...
    f_best_S21/1e9);
```

Output:








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PART B: OPTIMUM FREQUENCY ANALYSIS
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Minimum |S11| frequency = 10.1120 GHz
Maximum |S21| frequency = 10.1120 GHz

At optimum S11 frequency:
|S11| = 0.002134
|S21| = 0.944022
Return Loss = 53.4169 dB
Insertion Loss = 0.5004 dB

At 10 GHz:
Return Loss = 43.3379 dB
Insertion Loss = 0.5005 dB

FREQUENCY COMPARISON
====================
Reference frequency = 10 GHz
Optimum S11 frequency = 10.1120 GHz
Optimum S21 frequency = 10.1120 GHz
```